

THE EARTH



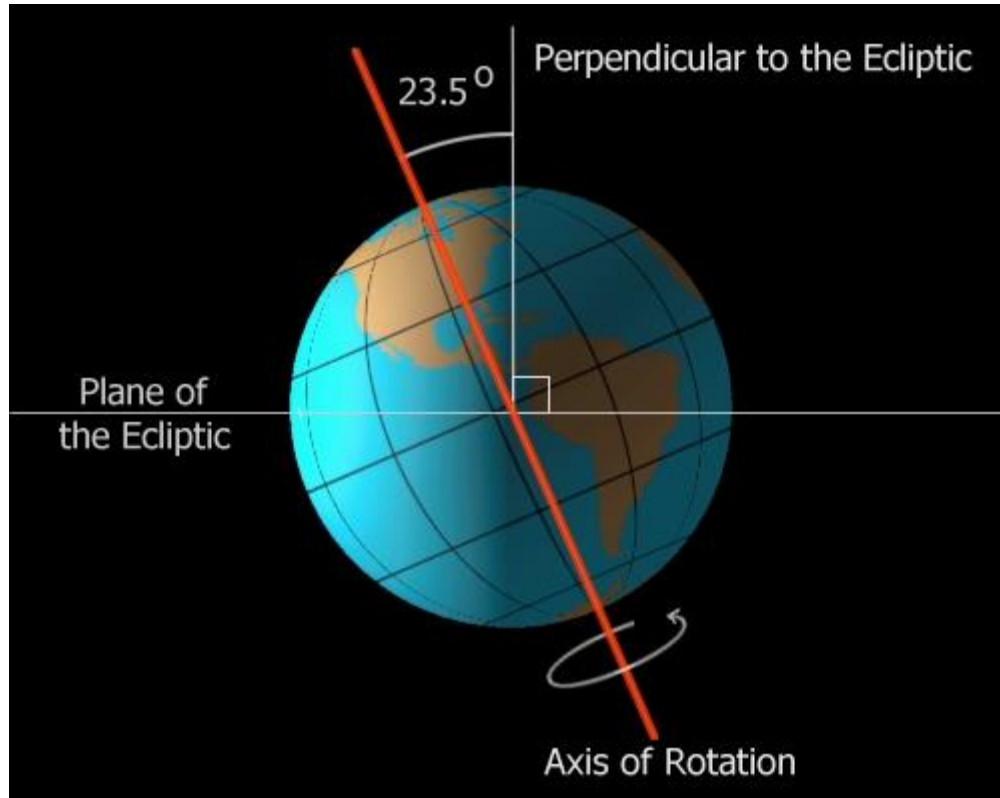
1. How did the Earth formed

A popular hypothesis for the formation of the earth is the **nebular hypothesis**. This idea dates back to the philosopher Immanuel Kant (1755) and Laplace (1796) and has been modified. Recent hypothesis starts with the solar system forming from a rotating, diffuse cloud of dust and gasses, 5 billion years ago.

How did the earth form in the condensing cloud? The earth was probably initially a homogeneous ball that heated from three sources: 1) energy of impacts, 2) gravitational compression, and 3) heat from radioactive disintegration.

2. Motion of the Earth

One of the important characteristics of the Earth is its movement. Like all planets in our Solar System, the Earth orbits or moves around the Sun. The Earth's axis is slightly tilted with respect to its orbit around the Sun, resulting in the change of seasons.



Earth's Axis Tilt

The Earth also follows the Sun in its motion through space. The various motions result in days, seasons and years.

2.1. *Rotates on its axis*

The Earth is a sphere that rotates on its axis that passes through the North and South poles. The time it takes for the Earth to make a complete rotation is approximately 24 hours (exactly 23.934 hours). This rotation results in daytime when an area is facing the Sun and nighttime when an area is facing away from the Sun.

The Earth



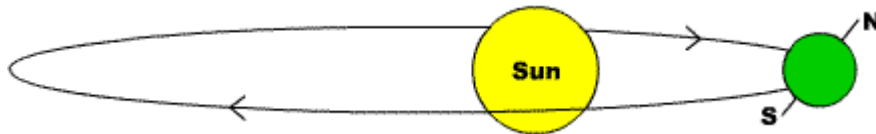
Earth rotates on its axis

Since we are on Earth, we do not sense its rotation, but we experience it by observing the relative motion of the Sun. It is similar to what you experience when you are sitting in a moving automobile and see the surroundings move by.

2.2. Revolves around the Sun

The Earth revolves around the Sun once every 365.25 days. Its velocity in space in its orbit is about 18.5 miles per second.

The shape of the orbit is an ellipse.



Earth rotates in an elliptical orbit around the Sun.

Leap years

Since a year has been designated as 365 days, a day is added every 4 years to even things out. That year is called a "Leap Year" and occurs on years divisible by 4, such as 2008. The extra day in a leap year is added in February.

Actually, the period is slightly less than 365.25 days. Thus a leap year is any year divisible by 4, except for years that are both divisible by 100 and not divisible by 400. Therefore, the year 2000 was a leap year, but the years 1700, 1800 and 1900 were not. That is complicated, but who wants things easy?

Seasons

The axis of the Earth is on a slight tilt with respect to the orbit around the Sun. That tilt is always in the same direction and results in the changing seasons on the Earth.

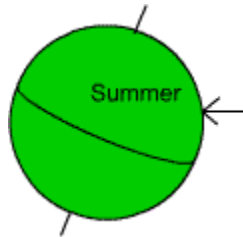


Tilt of the Earth is in the same direction

The reason the seasons change has to do with how direct the Sun is shining, as well as the length of a day.

Summer

When the Earth is in the part of the orbit around the Sun where the sunlight shines more directly on the Earth, the days are longer and less sunlight is reflected. That is summer, and the weather becomes warmer.



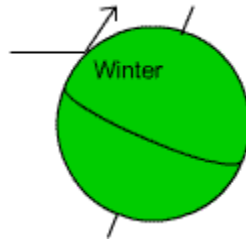
Sunlight shines more directly in summer

Due to the spherical shape of the Earth, there is more daylight in the summer the closer you get to the pole. One you cross the Arctic Circle in the Northern Hemisphere in summer or the Antarctic Circle in the summer in the Southern Hemisphere, a day can be 24 hours long. That is why they can it "the land of the midnight Sun."

Winter

On the other side of the orbit around the Sun, the angle of the sunlight is steeper, resulting in more light being reflected. Also, the days are shorter. This results in the colder weather of winter.

The Earth



Sunlight reflects off steeper angle in winter

Since the orbit of the Earth around the Sun is an ellipse, surprisingly, the Earth is closer to the Sun for a Northern Hemisphere winter than it is in the summer. That would imply that the Southern Hemisphere has slightly colder winters and warmer summers than the Northern.

2.3. Other motion

Besides spinning on its axis and rotating around the Sun, the Earth also follows the Sun's movement through the Milky Way galaxy.

The Milky Way, in turn, is moving through space, as part of the apparent expansion of the Universe. Its estimate velocity is thousands of miles per second depending on which galaxy you compare it.

3. Physical Properties of the Earth

3.1. Density of the Earth

The density of the Earth is higher than that of any other planet in our solar system: 5.43 g/cm^3 .

A scientist named Henry Cavendish is known for calculating the mass (and then density) of the Earth.

Since the Earth's mass is so great, it makes a gravity that compresses it more than every other kind of inner planet. The Earth hasn't all the huge "vapor envelopes" of the gas giants, so almost all of the matter on our planet consists of heavy solids. This is what makes Earth have the highest average density compared to any other planet in the solar system.

3.2. Acceleration due to gravity (g)

Is the acceleration that an object experiences because of gravity when it falls freely close to the surface of a massive body, such as a planet. It is also known as the *acceleration of free fall*.

In the case of the Earth, g comes out to be approximately 9.8 m/s^2 , though the exact value depends on location because of two main factors: the Earth's rotation and the Earth's equatorial bulge.

Why g varies from place to place?

The downward force of gravity is opposed by an outward centrifugal force due to the planet's rotation, which is greater at the equator than at higher latitudes. By itself, this effect would result in a range of values of g from 9.789 m/s^2 at the equator to 9.823 m/s^2 at the poles. This discrepancy is further accentuated because of the Earth's equatorial bulge, which causes objects at lower latitudes to be further from the planet's center than objects nearer the poles.

3.3. Pressure

The standard pressure of Earth at sea level is 101.3 KPa. However, different climatic conditions will change the air pressure of different regions. Temperature changes will cause changes in air pressure. As air temperature increases, it expands and its pressure will decrease. Some of the highest pressures were measured in Siberian winters. At low altitudes, pressures are generally high because the density of molecules above that area is high. The lowest point on earth, known as the Dead Sea is located more than 400 meters below sea level. The pressure of the atmosphere there is known to be extremely high

3.4. Heat

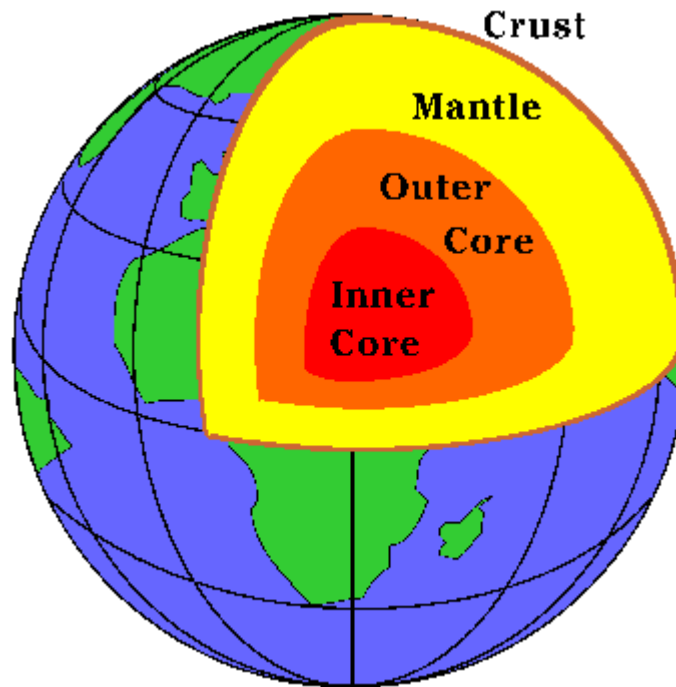
Imagine the earth as a big machine being run by two engines. Both are heat engines that convert heat into mechanical energy. One engine is external, the other internal.

The internal engine deep inside the earth is driven by heat moving from the hot interior. Moving plates, earthquakes and volcanoes are results of this engine. The internal engine is constantly running and has been for hundreds of millions of years.

The external engine is driven by solar power. Heat from Sun circulates the atmosphere and oceans. Water from the oceans evaporates and when it cools, it falls as rain or snow. When it falls on land the moisture helps break down rock and erodes the landscape. Mudflows and streams carry the particles away and deposit them lower. Mountains are carried away and reduced over long periods of time. Valleys are created in the same way.

Other physical properties of the Earth: electrical conductivity, magnetism, georadioactivity.

4. The internal structure of the Earth



4.1. The crust

The crust makes up only 0.5% of the Earth's total mass and can be subdivided into two main parts, continental and oceanic crusts. Both differ in thickness, density and composition. Although oceanic crust covers approximately 61% of the Earth's surface, it only comprises some 30% of the crustal mass, as the continental crust is much thicker.

Continental crust: from the surface of the Earth down to 30 km - 50 km. The exposed parts of the continental crust are less dense than oceanic crust (this is because oceanic crust contains minerals rich in heavier elements such as iron and magnesium). However, continental crust appears to be stratified (layered) and becomes denser with depth. It is largely composed of volcanic, sedimentary and granitic rocks, although the older areas are dominated by metamorphic rocks. The continental crust can be divided into:

1. **Stable continental regions. Thickness:** 35 km - 45 km
2. **Tectonically active regions** (e.g. the Andes, Himalayas). **Thickness:** 55 km - 70 km

Oceanic crust: from the surface of the Earth down to 10 km - 12 km. The oceanic crust can be divided into:

1. **Ocean basins**
2. **Mid-ocean ridges**
3. **Island arcs**

4.2. The Moho

The boundary between the crust and mantle is known as the Mohorovicic Discontinuity (Moho). It marks a significant change in chemical composition and is the boundary between the crust and mantle.

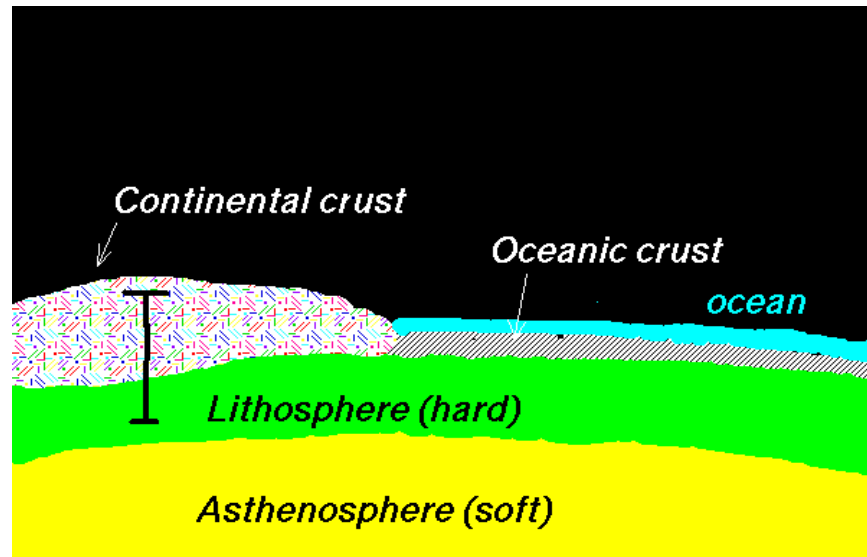
4.3. The mantle

The mantle is thought to be composed of rocks rich in magnesium and iron, and poor in silica. The mantle can be subdivided into **upper mantle**, **transition zone**, and **lower mantle**. There is also a zone in the upper mantle that we call the **low velocity zone** (located at depths of 70 km to 250 km) where scientists believe that most magmas (molten rock) are generated in this zone.

Upper mantle (measured from the base of the crust down to 400 km). 10 % of the Earth's total mass.

Transition zone (400 km - 1000 km below the Earth's surface). 17 % of the Earth's total mass.

Lower mantle (1000 km - 2900 km below the Earth's surface). 41 % of the Earth's total mass.

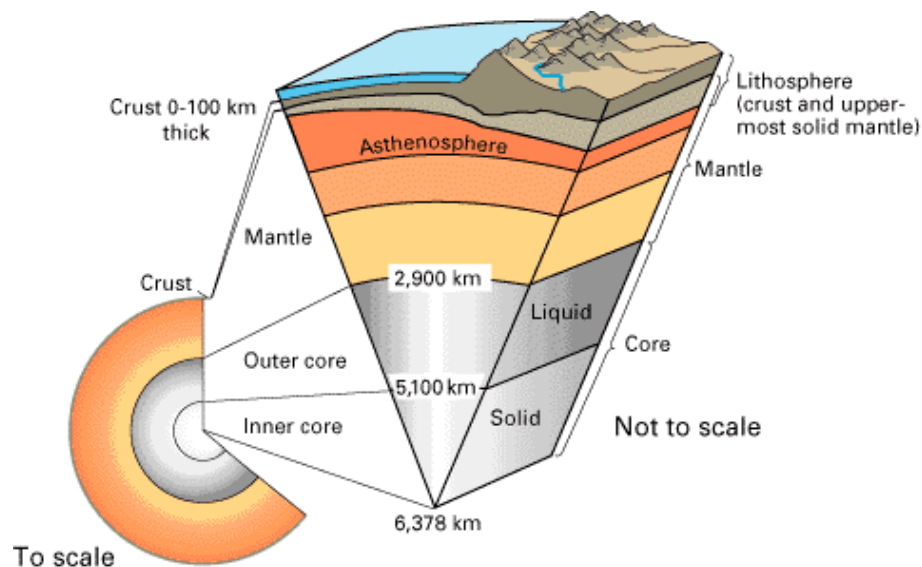


Continental crust, oceanic crust, lithosphere and asthenosphere

4.4. The core

The core it is believed to be composed primarily of a nickel-iron alloy (along with abundant platinum-group elements), consisting of a liquid outer zone, and a solid inner zone. It is also marked by an abrupt increase in pressure.

- **Outer core** (2900 km-5000 km below the Earth's surface). 30% of the Earth's mass.
- **Inner core** (5000 km-6370 km below the Earth's surface). 2% of the Earth's mass.



Earth's interior